Q1F | Predict Test Data

Class: CS P556 - Applied Machine Learning
Student: Reed Turgeon

Disclosure: This notebook was completed with the assistance of Claude Code (Opus 5.5) and the book *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd edition) by Aurélien Géron.

> F. Make predictions of the labels on the test data, using the trained model with chosen (best) hyperparameters. Summarize performance using the RMSE. Three teams with the lowest RMSE (who followed correctly all the previous steps) will receive bonus points.

Answer Summary

Predictions on the test data

- Both saved models predict every test row from the 8 raw attributes. Neither model has an imputer, which works because the test file has 0 missing values.

The trained models with the chosen hyperparameters

- D's linear regression and E's polynomial regression, each loaded exactly as its notebook selected, trained on the training set and saved.

RMSE

- E's polynomial model misses by less than D's linear model: 0.5024 against 0.5251 `Life Ladder` points.

Final submission

- E's polynomial regression (`polynomial_regression.joblib`), the saved model with the lower test RMSE.

Previous steps followed

- The test data is used once: nothing is refit or tuned after the test rows are scored.

Imports

In [1]:
```
import joblib
import pandas as pd
from sklearn.metrics import mean_squared_error

import Q1.consts as Q1_CONSTS

from utils import read_csv_to_df
```

```
pd.set_option('display.float_format', '{:.4f}'.format)
```

Fetch Data

In [2]:
```
test_df = read_csv_to_df(path=Q1_CONSTS.TEST_PATH)

ATTRIBUTES = [column for column in test_df.columns if column != Q1_CONSTS.TARGET]
X_test, y_test = test_df[ATTRIBUTES], test_df[Q1_CONSTS.TARGET]

print(f'{len(test_df)} test rows, {len(ATTRIBUTES)} attributes,
{test_df.isna().sum().sum()} missing values')
```

```
334 test rows, 8 attributes, 0 missing values
```

Method

- **Models:** the two models D and E selected and saved, loaded from `Q1/MODELS` as they are: D's linear regression (`linear_regression.joblib`) and E's polynomial regression (`polynomial_regression.joblib`). Each is one pipeline that takes the 8 raw attributes and builds its own columns, so nothing is fit on the test rows.
- **Metric:** RMSE in `Life Ladder` points on every test row, the metric D and E used.
- **Test data used once:** each model's RMSE is reported as it comes. Neither score is used to change or refit a model.
- **Submission:** the saved model with the lower test RMSE is the final submission.

Load the Models

In [3]:
```
MODELS = {  # the models D and E selected and saved, as they are
        'D: linear regression': joblib.load(Q1_CONSTS.MODEL_OUTPUT_DIR /
'linear_regression.joblib'),
        'E: polynomial regression': joblib.load(Q1_CONSTS.MODEL_OUTPUT_DIR /
'polynomial_regression.joblib'),
}
```

Test RMSE

In [4]:
```
PREDICTIONS = {name: model.predict(X_test) for name, model in MODELS.items()}

RMSE_COLUMN = f'test rmse ({Q1_CONSTS.TARGET_UNIT})'
RESULTS = pd.DataFrame({RMSE_COLUMN: {name: mean_squared_error(y_test, predicted,
squared=False) for name, predicted in PREDICTIONS.items()}})
RESULTS['submission'] = RESULTS.index == RESULTS[RMSE_COLUMN].idxmin()  # the lower test
RMSE is the final submission
RESULTS
```

Out[4]:

	test rmse (Life Ladder points)	submission
D: linear regression	0.5251	False
E: polynomial regression	0.5024	True

Notes / Observations

- E's polynomial model misses by less than D's linear model on new rows: 0.5024 against 0.5251 `Life Ladder` points.